Saudi Road Safety Analytics: An Academic Descriptive Report (2023–2024)

Data source: General Authority for Statistics (GASTAT), Road Transport Statistics 2024, official Arabic Excel workbook. Analysis: national annual indicators and regional serious road traffic accidents.

Abstract

This report examines changes in reported serious road traffic accidents, fatalities and injuries in Saudi Arabia between 2023 and 2024, and describes the geographical and within-/outside-city distribution of serious accidents across the 13 administrative regions in 2024. The study uses descriptive statistics, consistency checks and visual comparisons. Serious accidents increased by 4.22%, fatalities declined by 3.19%, and injuries rose by 0.31%. Makkah and Riyadh together accounted for 58.16% of reported serious accidents in 2024. These are counts and descriptive ratios; they are not estimates of exposure-adjusted risk or causal effects.

1. Research objectives and scope

The analysis addresses three questions: (i) how national serious-accident, fatality and injury counts changed from 2023 to 2024; (ii) how 2024 serious accidents were distributed across administrative regions; and (iii) how the reported accidents were divided between locations within and outside cities. The scope is intentionally descriptive; no prediction or causal inference is attempted.

2. Data provenance and verification

National indicators were drawn from the official GASTAT 2024 road-transport workbook (annual serious-accident totals in sheet 5-3 and 2024 fatalities and injuries in sheet 5-1). The 2024 regional accident table was verified against sheet 5-5 of the same Arabic workbook. An independent row-level audit compared three values for each of the 13 regions: within-city, outside-city and total accidents. All 39 comparisons matched exactly (39/39); the audit record is included with the project.

Data validation also confirmed that each regional total equals its within-city plus outside-city counts, that the 13 regional totals sum to the national 2024 count of 17,231, and that no region is duplicated. The main notebook embeds this verified snapshot for reproducibility and can rerun the workbook audit when the source file is available.

3. National changes, 2023–2024

Indicator	2023	2024	Change	Change (%)
Serious accidents	16,534	17,231	+697	+4.22%
Fatalities	4,423	4,282	−141	−3.19%
Injuries	24,002	24,077	+75	+0.31%

Percentage change is calculated as (2024 value − 2023 value) / 2023 value × 100. The opposite movements in serious accidents and fatalities show why each indicator should be interpreted separately rather than treated as a single safety measure.

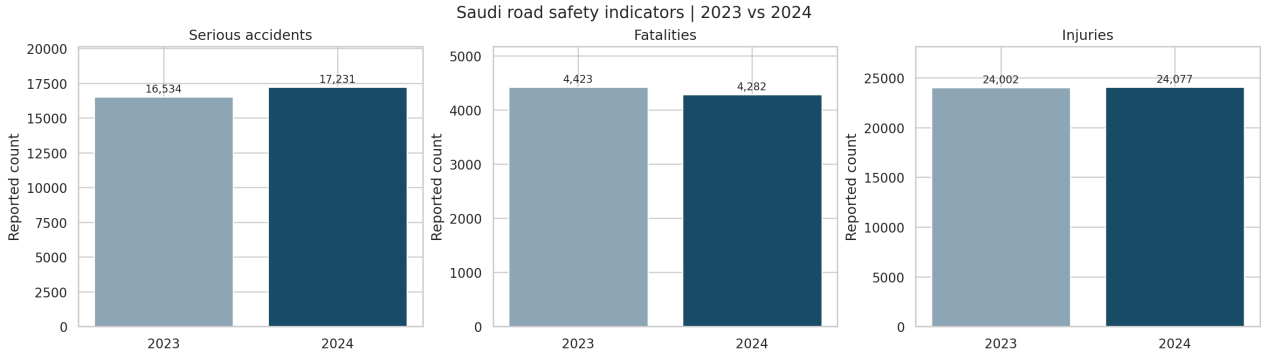


Figure 1. National road safety indicators in 2023 and 2024. Each panel uses its own scale; compare within an indicator, not bar heights across panels.

4. Descriptive severity ratios

For context, the analysis computes reported fatalities and injuries per 100 serious accidents. In 2023, these ratios were 26.75 fatalities and 145.17 injuries per 100 serious accidents; in 2024, they were 24.85 and 139.73, respectively. These ratios are descriptive quotients of aggregate counts. They do not represent the probability of death or injury in an individual crash, and they should not be interpreted as clinical or causal severity measures.

5. Regional distribution of serious accidents (2024)

The 13 regions reported 17,231 serious road traffic accidents in total. Makkah recorded 5,825 accidents (33.81% of the national total), followed by Riyadh with 4,196 (24.35%) and Madinah with 1,777 (10.31%). Together, the three regions accounted for 68.47% of reported serious accidents. These figures describe where recorded events were concentrated, not which region had the highest exposure-adjusted risk.

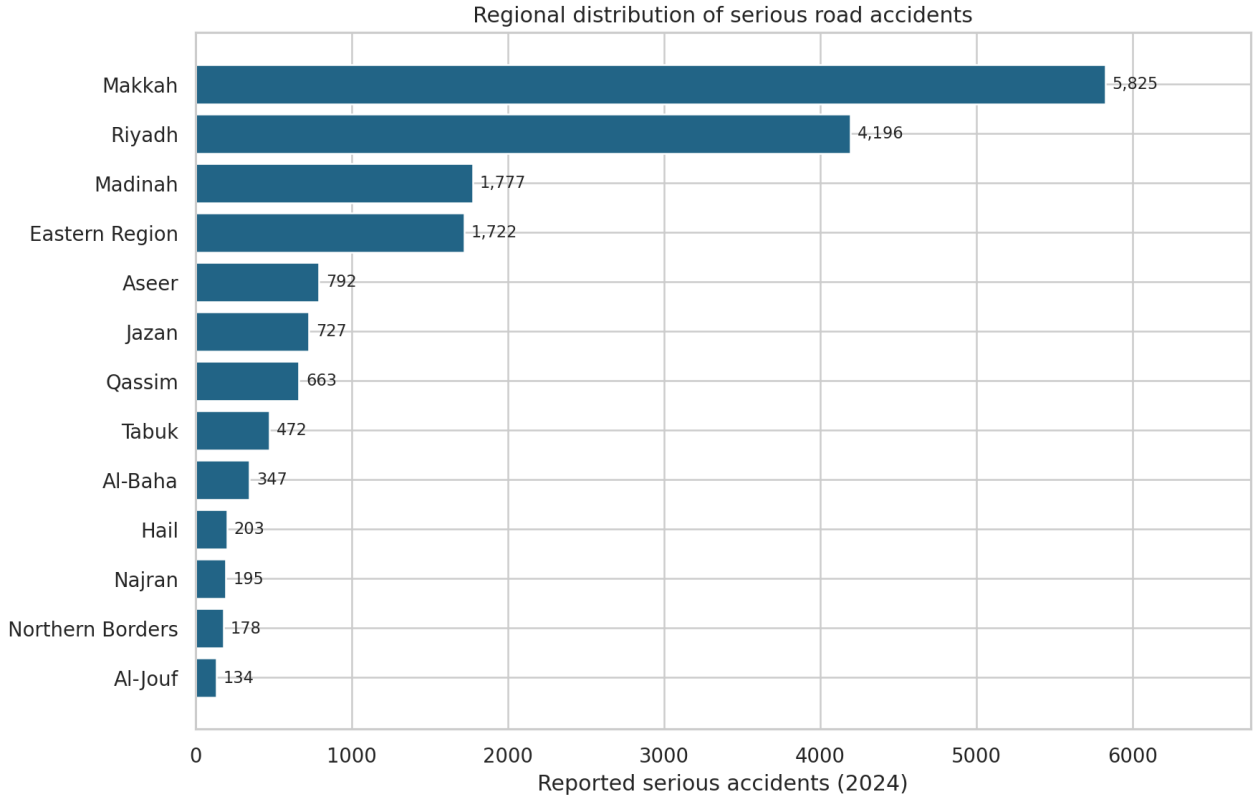


Figure 2. Number of serious road traffic accidents by administrative region, 2024. Counts are not adjusted for population, traffic volume or distance traveled.

6. Accidents within and outside cities

Of the 17,231 reported serious accidents in 2024, 10,392 (60.31%) occurred within cities and 6,839 (39.69%) occurred outside cities. The distribution differs by region: for example, Aseer recorded 688 of its 792 serious accidents outside cities (86.87%), whereas Riyadh recorded 3,150 of 4,196 within cities (75.07%). The percentages describe regional composition rather than risk or exposure.

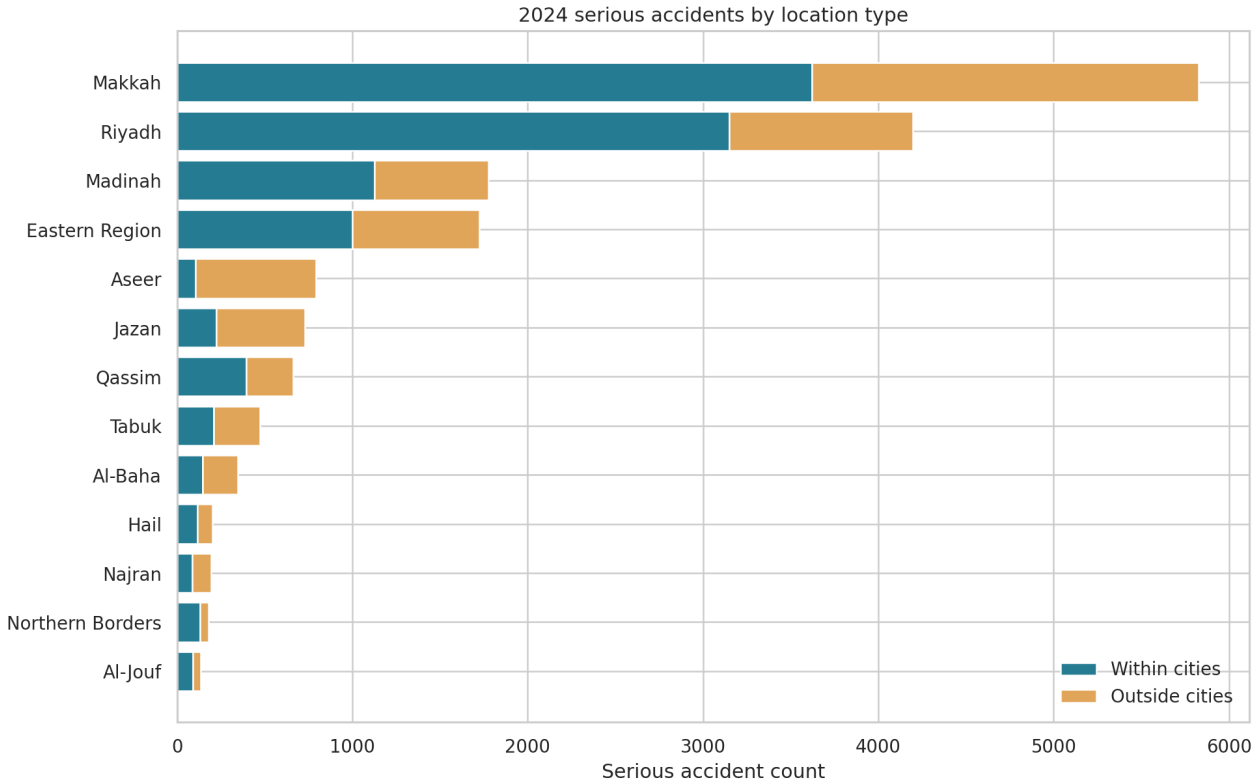


Figure 3. Within-city and outside-city serious accident counts by region. Stacked lengths show each region’s absolute reported total.

7. Composition and interpretation

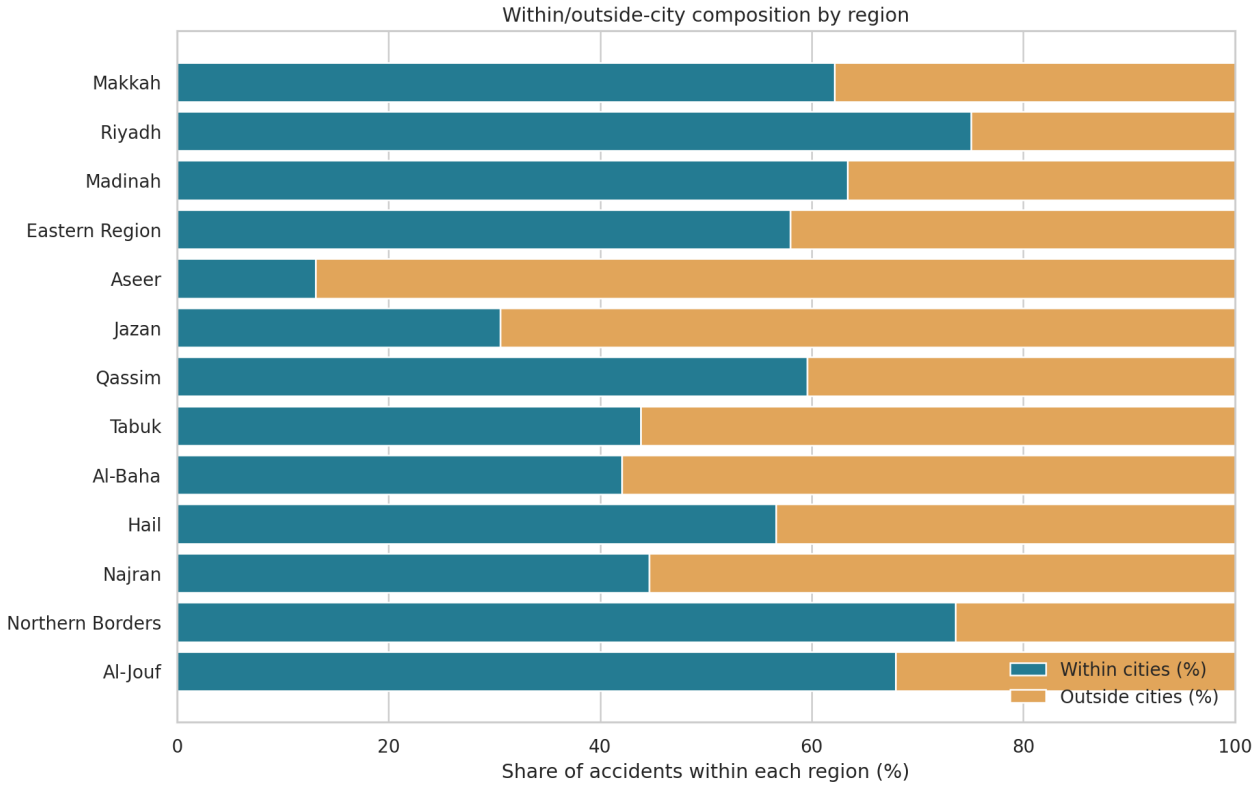


Figure 4. Within-city versus outside-city share within each region. Unlike Figure 3, each region is normalized to 100%.

The count chart and proportional chart answer different questions. Figure 3 highlights the absolute contribution of each region to national accident totals, while Figure 4 compares the relative within-/outside-city composition within a region. A high outside-city percentage does not imply that outside-city roads are intrinsically more dangerous: traffic exposure, road type, travel distances, enforcement, and reporting practices may differ substantially.

8. Analytical limitations

First, the national comparison covers only two years and cannot establish a long-term trend. Second, the regional analysis uses aggregated administrative counts without denominators such as vehicle-kilometers traveled, population or traffic volume; it cannot rank regions by true road-traffic risk. Third, the within-/outside-city classification does not isolate road design, crash mechanism, time of day or individual-level outcomes. Finally, observational aggregate data cannot identify causal drivers of the observed differences.

9. Conclusion

The official data show a 4.22% increase in reported serious accidents from 2023 to 2024, alongside a 3.19% decrease in fatalities and a modest 0.31% increase in injuries. In 2024, recorded serious accidents were geographically concentrated, with Makkah, Riyadh and Madinah accounting for more than two-thirds of the national total. Within-city accidents represented 60.31% nationally, although the composition varied considerably across regions. These findings provide a reproducible descriptive baseline for further road-safety investigation; evaluating comparative risk would require suitable exposure data and more detailed records.

References and reproducibility

General Authority for Statistics (GASTAT). Road Transport Statistics 2024. Official Arabic Excel workbook, sheets 5-1, 5-3 and 5-5. Website: https://www.stats.gov.sa/

Supporting materials: Saudi_Road_Safety_Analytics_Polished_FINAL.ipynb; Saudi_Road_Safety_Official_Regional_Audit.csv; and four exported figures. All numerical claims in this report are based on the included verified project dataset and audit.